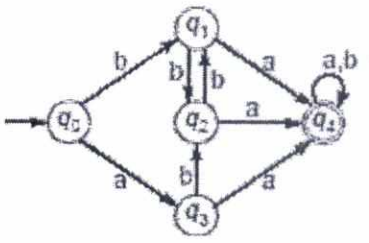
Roll Number: _____	
Thapar Institute of Engineering and Technology, Patiala	
Department of Computer Science and Engineering **Auxiliary Examination**	
BE COE & CSE (Third Year)	**UCS701: Theory of Computation**
17th August 2024, 5:30 PM	**Time: 3Hours, Max Marks:100**
Name of Faculty: Sunita Garhwal	

Note: Attempt all questions (subparts) in sequence at one place. Assume missing data, if, any, suitably.

Q1.	Given L_1={x|00 is not substring of x, x ∈ {0,1}*} L_2={x| x ends with 01, x ∈ {0,1}*} Draw the Finite Automata recognizing the following languages: a) $L_1 \cap L_2$ b) $L_1 \cup L_2$ c) $L_1 - L_2$	 4 3 3
Q2.	Given the regular expression $r = 11(0+1)^*101$ over the alphabet $\sum$={0,1} a) Convert the given r into NFA using Thompson's construction. b) Convert the obtained NFA into DFA using subset construction. c) Minimize the obtained DFA in 2(b).	 5 5 5
Q3.	Consider following context free grammar: $A \rightarrow BAB \mid B \mid \varepsilon$ $B \rightarrow 00 \mid \varepsilon$ Convert the above grammar into Chomsky Normal Form(CNF).	10
Q4.	Draw the flowchart for the language $L=\{a^n b^m c^m d^n \mid m,n \geq 1\}$ accepted by the push down machine. Construct the transition diagram for the above designed flowchart.	10
Q5.	Write down the logic for the design of a Turing machine to accept the language $L = \{0^n 1^n \mid n \geq 0\}$. Design the Turing machine for the above designed logic.	15
Q6.	Convert the following into minimum DFA.  	10
Q7.	Given the Context-free grammar G: S->AS/SB/0/1 A->AA/0 B->BB/1 Apply CYK algorithm to determine whether the string w=100100 ∈L(G) or not?	10
Q8.	Construct a Moore machine that takes binary number as input and produces residue '5' as output? Convert it into an equivalent Mealy machine.	10

[P.T.O.]

Q9.	Using Pumping Lemma, Prove that $L = \{0^i 1^i \mid i \geq 0\}$ over alphabet $\sum=\{0,1\}$ is not regular.	10